For categories, weak equivalences and equivalences of precategories coincide

The Univalent Foundations Program

hott-book:categories.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'ct:catweq' (source lines 658-661, Lemma 9.4.7); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof use no label from outside the retained range.

Editorial note. The retained counter values (chapter 9, section 4, theorem offset 6) make the counted lemma print as Lemma 9.4.7, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: for categories, weak equivalences are equivalences of precategories (Lemma 9.4.7).

Allowed standard prerequisites (editorial).

- Isomorphism of objects versus identity. In a category, the type a = a' is equivalent to the type of isomorphisms a ∼ a', and this equivalence is respected by the action of a functor.. It is used to turn the isomorphism g : a ∼ a' into a path and then to transport along it at source line 666.
- Characterisation of equalities in dependent pair types. Two elements of a sigma type are equal as soon as their first components are equal and the second are identified after transport.. It is the final step at source line 667.
- Weak equivalence. A functor is a weak equivalence when it is fully faithful and essentially surjective, i.e. when essential surjectivity is only asserted with a truncation.. It is the notion compared with being an equivalence of precategories.

Lemma 9.4.7. *If $F : A \to B$ is fully faithful and A is a category, then for any $b : B$ the type $\sum_{(a:A)}(Fa \cong b)$ is a mere proposition. Hence a functor between categories is an equivalence if and only if it is a weak equivalence.*

Proof. Suppose given (a, f) and (a', f') in $\sum_{(a:A)}(Fa \cong b)$. Then ${f'}^{-1} \circ f$ is an isomorphism $Fa \cong Fa'$. Since F is fully faithful, we have $g : a \cong a'$ with

$Fg = {f'}^{-1} \circ f$. And since A is a category, we have $p : a = a'$ with $\mathsf{idtoiso}(p) = g$. Now $Fg = {f'}^{-1} \circ f$ implies $((F_0)(p))_*(f) = f'$, hence (by the characterization of equalities in dependent pair types) $(a, f) = (a', f')$.

Thus, for fully faithful functors whose domain is a category, essential surjectivity is equivalent to split essential surjectivity, and so being a weak equivalence is equivalent to being an equivalence. □